

K-Map

If 'n' is number of input variables, then total combinations = 2^n and total logical expressions = 2^{2^n}

eg. $n=2$ a, b

Possible logical expressions

$$= 2^{2^2} = 2^4 = \boxed{16}$$

ab	$a+b$	0	1
$a\bar{b}$	$a+\bar{b}$	a	$\bar{a}$
$\bar{a}b$	$\bar{a}+b$	b	$\bar{b}$
$\bar{a}\bar{b}$	$\bar{a}+\bar{b}$	$a \odot b$	$a \oplus b$

Positional wt. 2^i ... 32 16 8 4 2 1

• Binary To Gray Code Conversion

MSB (Most significant bit) (left most)

$a_3 \ a_2 \ a_1 \ a_0$

↓

G_3

$$G_3 = a_3$$

$$G_2 = a_3 \oplus a_2 \quad \text{Ex-OR}$$

$$G_1 = a_2 \oplus a_1$$

$$G_0 = a_1 \oplus a_0$$

eg.:

bin	Gray
0 0 0 →	0 0 0
0 0 1 →	0 0 1
0 1 0 →	0 1 1
0 1 1 →	0 1 0
1 0 0 →	1 1 0
1 0 1 →	1 1 1
1 1 0 →	1 0 1
1 1 1 →	1 0 0

Gray code also called unit distance code.
 Change of 1 bit in each stage.

K-Map is a graphical representation through which we minimize any boolean function.

when variable increase from 3, 4, 5 boolean laws and methods gets complicated to find minimized expression.

K-Map uses Gray code or unit distance code.

Total number of cells / squares = 2^n , $n \rightarrow$ no. of variables

ex $Y = \sum m(1, 2, 8, 11, 13)$

4 bits are required to represent 13 in binary. So, its a 4 variable K-map.

Adjacent cells: From going one cell to another, if only one variable changes. These two cells are called adjacent cells.

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

$$\begin{matrix} A & B & C \\ 1 & 1 & 0 = 6 \\ 4 & 2 & 1 \end{matrix} 2^i$$

We make grouping of adjacent cells in 2^i form $i=0, 1, \dots$ and in this i variables are eliminated.

For example, if $i=3$ means $2^3 = 8$. In an octet, 3 variables will be eliminated.

Prime Implicants (PI)

It is a group of minterms or minterm.

**** Essential Prime Implicant (EPI)

It is prime implicant in which there is atleast one minterm, which is not involved in any other group. Final minimized expression will be sum of EPI's.

Ques Minimize following boolean funcⁿ using K-Map. Also determine total no. of PI and EPI.

(i) $Y = \sum m(1, 3, 6, 7)$

		$\bar{B}\bar{C}$		$\bar{B}C$		BC		$B\bar{C}$	
		00		01		11		10	
$\bar{A}$	BC	0	0	1	1	1	1	0	0
	A	0	0	1	1	1	1	0	0
A	BC	0	0	0	0	1	1	0	0
	A	1	1	0	0	1	1	0	0

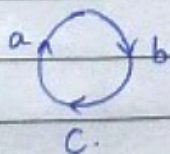
$PI_1 \rightarrow EPI \quad \bar{A}\bar{C}$

$PI_2 \rightarrow X$

$Y = \bar{A}\bar{C} + AB$

$PI_3 \Rightarrow EPI \quad AB$

Consensus Theorem: (Redundancy theorem)



Cyclic order SOP $AB + BC + \bar{C}\bar{A} = AB + \bar{A}\bar{C}$

POS $(A+B) \cdot (B+C) \cdot (C+\bar{A}) = (A+B) \cdot (\bar{A}+C)$

$\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{C}\bar{A} = \bar{A}\bar{B} + \bar{A}\bar{C}$

4-Variable K-map.

CD		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
		00	01	11	10
AB	$\bar{A}\bar{B}$	0	1	1	1
	$\bar{A}B$	0	1	1	1
AB	$A\bar{B}$	0	1	1	1
	AB	0	1	1	1

$$A B C D$$

$$1 \ 1 \ 0 \ 1$$

$$8 \ 4 \ 2 \ 1$$

$$= 13$$

(i) $Y = \sum m(0, 1, 2, 3, 8, 9, 10)$

$$\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{B}\bar{D}$$

(iii) $Y = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

CD		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
		00	01	11	10
AB	$\bar{A}\bar{B}$		1		
	$\bar{A}B$		1	1	1
AB	$A\bar{B}$	1	1	1	1
	AB	1	1	1	1

$$PJ_5 \rightarrow EPI X$$

all 1's are covered by other groups

$$Y = \bar{A}\bar{C}D + \bar{A}BC + ACD + AB\bar{C}$$

Solar

(iv) $Y = \prod M(0, 1; 2, 3, 4, 5, 8, 9, 10, 12, 13)$

		CD				$\bar{C}\bar{D}$				$\bar{C}D$				CD			
		CD		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$\bar{C}\bar{D}$	
A	B	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\bar{B}$	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\bar{A}$	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\bar{B}$	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$\bar{A}$	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$$Y = \bar{C} + AB + BD = C \cdot (A+B) \cdot (B+D)$$

We need to always perform operations on K-Map using POS (maxterm).

Because PI and EPI laws are applied on minterm only.

		$\bar{C}\bar{D}$				$\bar{C}D$				CD				$C\bar{D}$			
		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$\bar{C}\bar{D}$		$\bar{C}D$	
A	B	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0
$\bar{A}$	0																
$\bar{B}$	0																
$\bar{A}$	0																
B	1																
A	1																
B	1																
A	1																
$\bar{B}$	0																

$$Y = ACD + BC$$

Distributive law

~~BC~~

$$C (AD + B)$$

$$C (B + A)(B + D)$$

To convert SOP into POS or vice-versa, apply distributive law, i.e.,
 $X + YZ = (X + Y) \cdot (X + Z)$
 $X \cdot (Y + Z) = XY + XZ$

Q Minimize the following boolean funcⁿ and implement minimized boolean funcⁿ using ex-OR gate only.

i) $Y = \sum m(0, 5, 8, 10, 13, 15) + \sum d(1, 2, 3)$

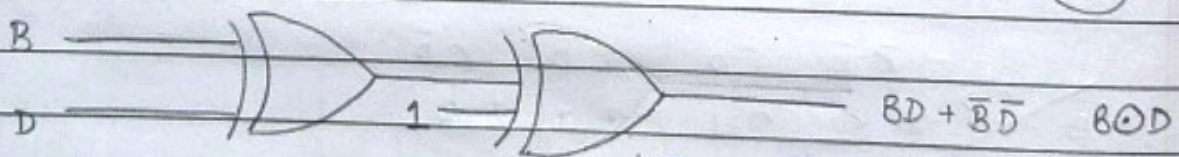
CD.		$\bar{C} \bar{D}$		$\bar{C} D$		$C \bar{D}$		$C D$	
AB		0	0	0	1	1	1	1	0
$\bar{A}$	$\bar{B}$	0	X	X		X		X	
$\bar{A}$	B	0		1		1			
A	$\bar{B}$	1		1		1			
A	B	1							
$\bar{A}$	$\bar{B}$	0	1					1	

If 'd' is used in a group, its value becomes 1.
 If 'd' is not used in group, its value is 0.

$$Y = BD + \bar{B}\bar{D}$$

$$A\bar{B} + \bar{A}B = A \times 0 + \bar{A} \times 1$$

$$B = 1 = \bar{A}$$



$$a \odot 0 = \bar{a}$$

$$a \odot 1 = a$$

$$a \odot \bar{a} = 0$$

$$a \odot a = 1$$

$$a \oplus 1 = \bar{a} \text{ (NOT)}$$

$$a \oplus 0 = a \text{ (Buffer)}$$

$$a \oplus \bar{a} = 1$$

$$a \oplus a = 0$$

(iv) $Y = \sum m(3, 7, 10, 11, 15) + \sum d(0, 1, 2, 5, 8, 14)$

		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$C\bar{D}$	
AB	CD	00	01	11	10	00	01	11	10
	00	(X)	(X)	(I)	(X)				
01			(X)	(I)					
11				(I)	(X)				
10		(X)		(I)	(I)				

$Y = CD + AC \quad (\text{III} + \text{I}) \checkmark$
 $Y = CD + \bar{B}\bar{D} \quad (\text{IV} + \text{II})$
 $Y = AC + \bar{A}D \quad (\text{I} + \text{IV})$
 $Y = CD + \bar{B}C \quad (\text{III} + \text{V}) \checkmark$

		$\bar{C}\bar{D}$		$\bar{C}D$		CD		$C\bar{D}$	
AB	CD	00	01	11	10	00	01	11	10
	00	(X)	(X)			(X)			
01		(X)	(X)			(X)			
11									
10									

$Y = C \cdot (A + D) = AC + CD \quad \checkmark \quad \text{I} + \text{II}$
 $Y = C \cdot (\bar{B} + D) = CD + \bar{B}C \quad \checkmark \quad \text{III} + \text{I}$
 $Y = (\bar{A} + C) (\bar{B} + D) = \bar{A}\bar{B} + \bar{A}D + \bar{B}C + CD \quad \text{III} + \text{IV}$
 $Y = (\bar{A} + C) (A + D) = A\bar{A} + \bar{A}D + AC + CD \quad \text{II} + \text{IV}$
 $= \text{Consensus theorem.}$
 $= AC + \bar{A}D.$

$Y = (C + \bar{D}) (\bar{B} + D) \quad \text{III} + \text{IV}$
 $= \bar{B}C + CD + \bar{B}\bar{D} + D\bar{D}$
 $=$

Minimize following boolean function.

CD		00 01 11 10			
AB	00	1	1	1	d
	01	1	d		
	11	1	d		1
	10	1	1	d	1
	00				

$$Y = \bar{B} + \bar{C} + A\bar{D} = \bar{B} + \bar{C} + A\bar{D}$$

5- Variable K-Map

Total cells = $2^5 = 32$.

We construct 2 separate tables with 16 cells in each.

A	B	C	D	E	
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	1
0	1	1	1	1	15

A	B	C	D	E	
1	1	1	1	1	31
1	1	1	1	0	14
1	1	1	0	1	13
1	1	0	1	1	11
1	1	0	1	0	9

A	DE			
	BC	00	01	11
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

BC	DE			
	00	01	11	10
	16	17	19	18
	20	21	23	22
	28	29	31	30
	24	25	27	26

$$\begin{aligned} &A B C D E \\ &0 1 1 1 0 \\ &8 4 2 1 \\ &= 14 \end{aligned}$$

$$\begin{aligned} &A B C D E \\ &1 0 0 0 1 \\ &16 8 4 2 1 = 17 \end{aligned}$$

Adjacent cells = $a_{ij} = b_{ij}$

Q Minimize given boolean function:

(i) $Y = \sum m(5, 7, 9, 11, 13, 15, 21, 23, 27, 29, 31)$

$\bar{A}$	DE	00	01	11	10
Bc	00				
	01		1	1	
	11		1	1	
	10		1	1	

A	DE	00	01	11	10
Bc	00				
	01		1	1	
	11		1	1	
	10		1	1	

both quad are in same position (adjacent cell)
Hence, it forms an octet O_1 ,

q_2

$$Y = O_1 + q_1 + q_2$$

$$Y = CE + BDE + \bar{A}BE$$

(ii) $Y = \sum m($

$\bar{A}$	DE	00	01	11	10
Bc	00		X	X	
	01	1	X	1	1
	11	X	1	X	
	10	X	1	1	X

A	DE	00	01	11	10
Bc	00		X		
	01	1	1		1
	11		1	X	
	10	X	X	1	1

O_2

O_1

O_3

$$Y = O_2 + O_3 + O_1 + q_1$$

$$Y = \bar{A}E + B\bar{C} + \bar{D}E + B\bar{C}\bar{E}$$

$\bar{A}$	BC	DE	00	01	11	10
0	0		X	1	1	X
0	0					
0	1			X	1	
1	0				1	
1	1				1	
1	0				1	

A	BC	DE	00	01	11	10
0	0		X	1	1	X
0	0					
0	1				X	
1	0				1	X
1	1		X	X	1	X

No EP1

Q_1 Q_2

$$Y = Q_1 + Q_2$$

$$= DE + \bar{B}\bar{C}$$